

Portfolio for Data Science Com618 (Stroke Risk Detection)

Dafe Edesiri Otudje 102613665/1

Introduction

Stroke is the second leading cause of death globally, with approximately 80% being preventable through early intervention (WHO, 2023; Feign et al., 2022). Machine learning offers promising approaches for identifying at risk individuals, enabling targeted preventive measures. However, stroke prediction faces a critical challenge: severe class imbalance in real world datasets where stroke occurrence represents only 4-5% of cases (Melnykova et al., 2025). This imbalance creates a “Accuracy Paradox” where models may achieve high overall accuracy while failing to identify the high-risk patients who require intervention most.

Recent literature demonstrates varied success in addressing these challenges. Gupta et al. (2024) reported F1-scores of 92% using neural networks on the Healthcare Stroke Dataset, while Melnykova et al. (2025) achieved 90% F1-scores using SMOTE-based optimized Random Forest classifiers. However, these results often stem from balanced experimental conditions or synthetic test sets that may not reflect natural disease prevalence. Questions remain regarding which imbalance handling strategies, synthetic oversampling (SMOTE) versus cost-sensitive learning (class-weight adjustment) provides more reliable and generalizable performance in realistic clinical scenarios.

Research Question: Which class imbalance handling approach provides more reliable stroke risk prediction when evaluated on realistic, imbalanced test data reflecting actual disease prevalence?

This study analyzes the Healthcare Stroke Prediction Dataset (Kaggle, 2021; 5,110 patients, 12 variables) with three objectives: (1) conduct rigorous preprocessing and exploratory analysis to understand the key drivers of stroke risk; (2) compare SMOTE-based oversampling with cost-sensitive learning methods across multiple algorithms (Logistic Regression, Random Forest, XGBoost); and (3) identify clinically meaningful patient subgroups through unsupervised clustering. Our approach prioritizes Recall over Precision, accepting a higher false-positive rate to minimize the clinical risk of missed diagnoses, a strategy aligned with the “Safety-First” imperative of preventative healthcare.

Methods

Dataset

Source: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>

This study utilizes the HealthCare Stroke Prediction Dataset sourced from Kaggle (Fedesoriano, 2021), selected for its comprehensive coverage of clinically relevant risk factors and realistic representation of stroke prevalence in population level screening scenarios. The dataset comprises 5,110 patient records across 12 attributes, including demographic variables (age, gender, marital status, residence type), clinical measurements (average glucose level, body mass index), medical history (hypertension, heart disease), and lifestyle factors (work type, smoking status). The target variable indicates stroke occurrence (binary: 0/10).

Three criteria guided the selection of the dataset: (1) clinical relevance, all variables match recognised stroke risk factors reported in medical literature (AHA, 2023); (2) realistic class distribution, the dataset shows 4.87% stroke prevalence (249 positive cases), reflecting real-world disease incidence where rare event prediction presents real methodological challenges; and (3) data quality, minimal missing values (3.9% BMI, 30.2% smoking status) with no duplicate records. A detailed breakdown of dataset attributes is given in Table 1.

Attribute	Type	Values/Range	Description	Missing (%)
id	Integer	1-72940	Unique patient identifier	0.0
gender	Categorical	Male, Female, Other	Patient gender	0.0
age	Continuous	0.08-82.0 years	Patient age	0.0
hypertension	Binary	0, 1	History of hypertension	0.0
heart_disease	Binary	0, 1	History of heart disease	0.0
ever_married	Categorical	Yes, No	Marital status	0.0
work_type	Categorical	Private, Self-employed, Govt_job, children, Never_worked	Employment category	0.0
Residence_type	Categorical	Urban, Rural	Residence location	0.0
avg_glucose_level	Continuous	55.12-271.74 mg/dL	Average blood glucose	0.0
bmi	Continuous	10.3-97.6	Body Mass Index	3.9
smoking_status	Categorical	formerly smoked, never smoked, smokes, Unknown	Smoking history	30.2
stroke	Binary	0, 1	Stroke occurrence (target)	0.0

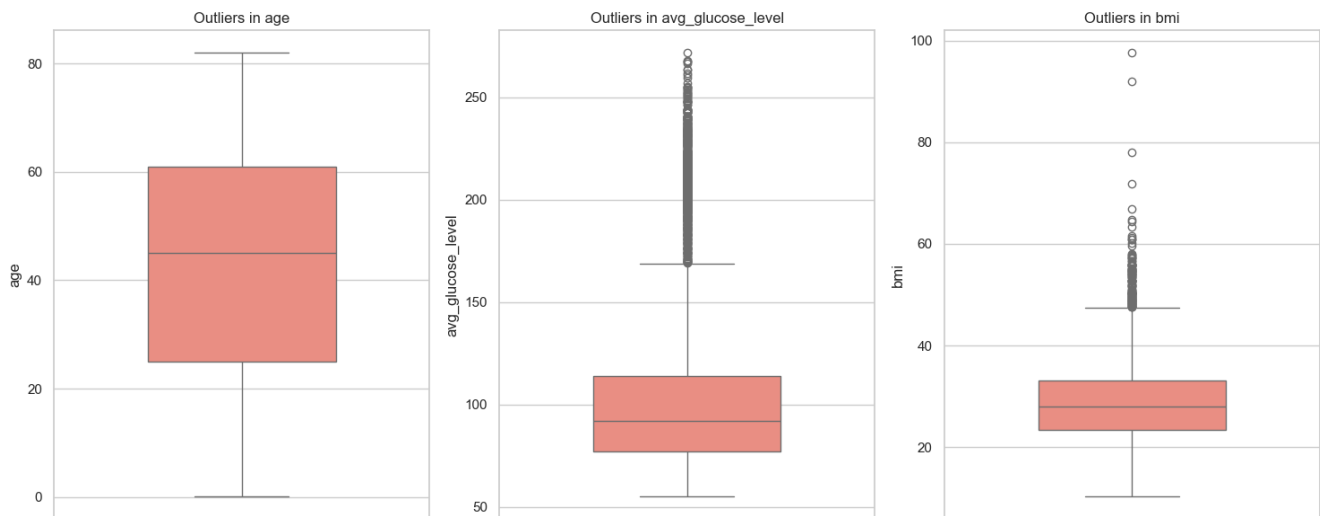
Dataset Statistics: 5,110 total records | 249 stroke cases (4.87%) | 4,861 non-stroke cases (95.13%)

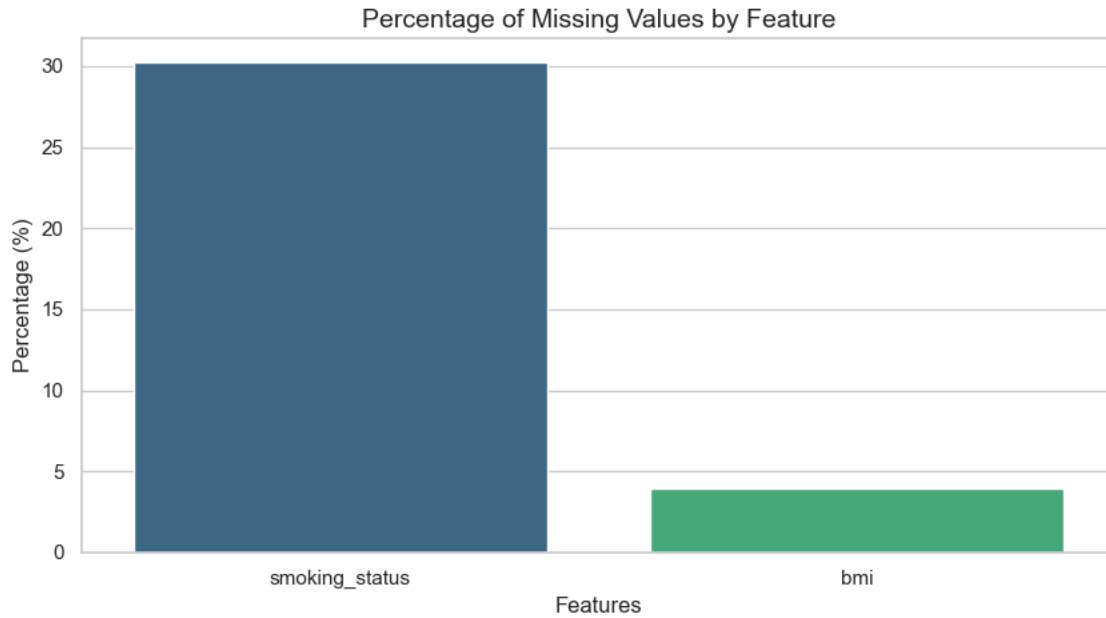
Analysis and Results

Data Quality Assessment

Initial data quality assessment revealed two primary issues requiring intervention. First, the `smoking_status` variable contained 1,544 missing values (30.2% of records), exceeding the 20% threshold recommended by Schafer (1999) for safe imputation. Second, `bmi` exhibited 201 missing values (3.9% of records).

Given that continuous variables are susceptible to measurement errors and extreme physiological values, we applied the Interquartile Range (IQR) method to detect outliers in age, average glucose level, and BMI, the three continuous numerical features most likely to contain extreme values requiring clinical validation. This analysis identified 627 glucose outliers (12.27%) and 110 BMI outliers (2.15%) exceeding typical bounds, while age showed no outliers (Figure 1).





```
print("\n" + "=" * 70)
print("OUTLIER DETECTION (IQR Method)")
print("=" * 70)

numerical_cols = ['age', 'avg_glucose_level', 'bmi']

# Visualization: Boxplots for Outliers
plt.figure(figsize=(15, 6))
for i, col in enumerate(numerical_cols):
    plt.subplot(*args: 1, 3, i + 1)
    sns.boxplot(y=df[col], color='salmon', width=0.5)
    plt.title(label: f'Outliers in {col}', fontsize=12)

    # Calculate IQR for the printed report
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

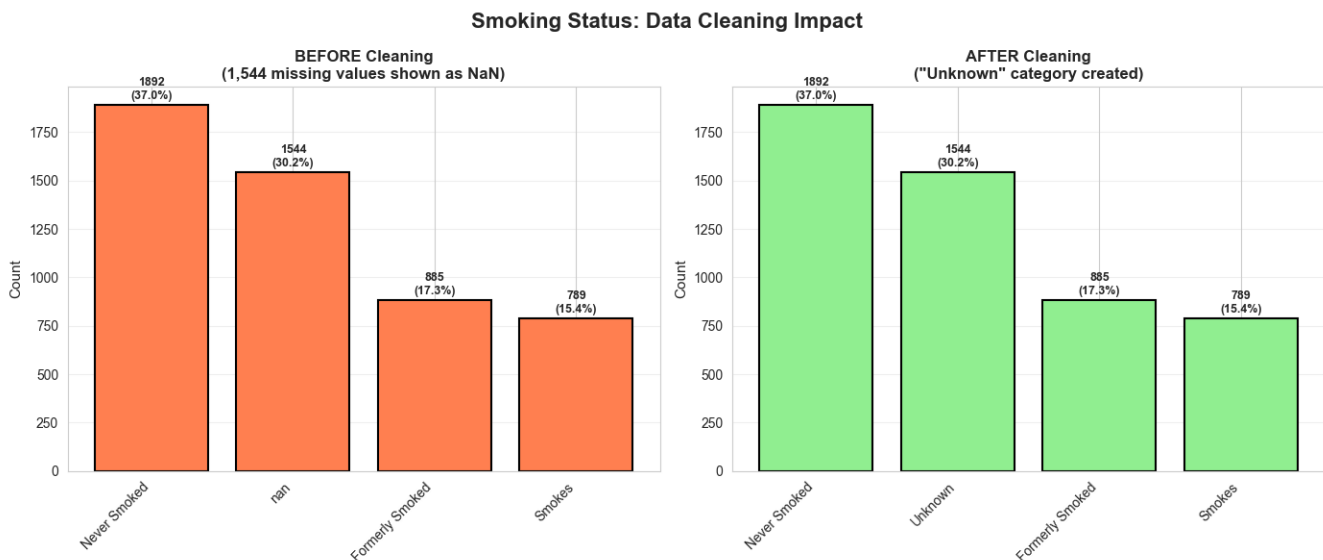
    print(f"\n{col}:")
    print(f"  Lower Bound: {lower_bound:.2f} | Upper Bound: {upper_bound:.2f}")
    print(f"  Outliers: {len(outliers)} ({(len(outliers) / len(df) * 100):.2f}%)")

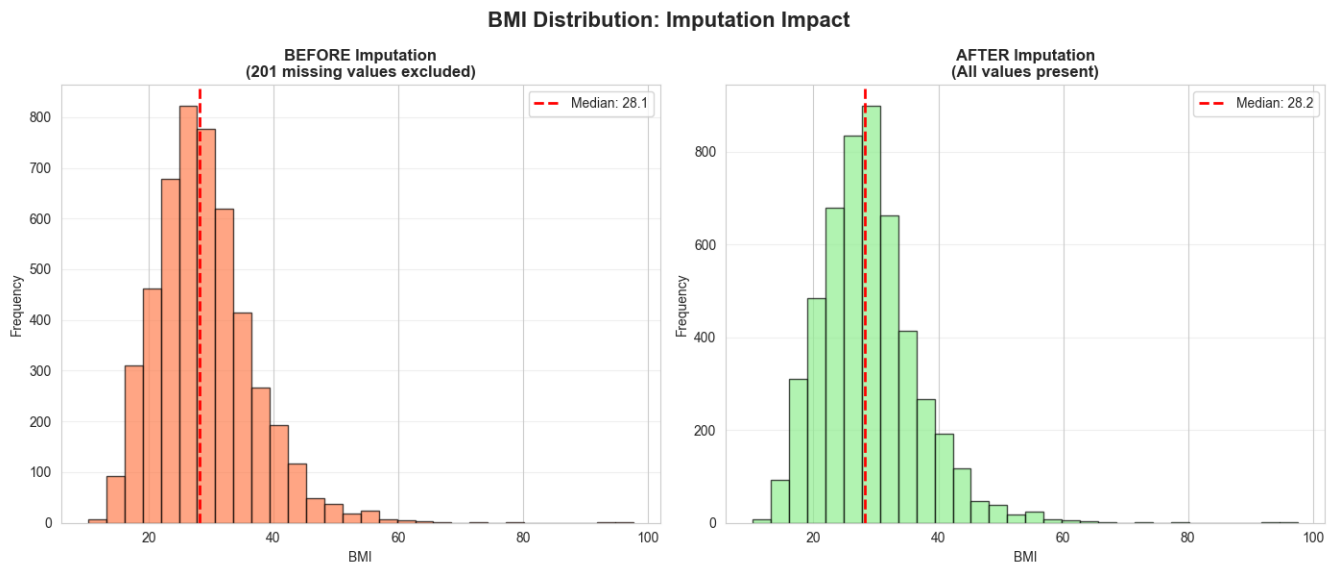
plt.tight_layout()
plt.savefig('outlier_boxplots.png')
print("-> Visualization saved: outlier_boxplots.png")
```

Data Cleaning Decisions

To maintain data integrity while acknowledging this limitation, I created an “Unknown” category for smoking_status due to the 30.2% missing rather than imputing values, preserving data integrity while acknowledging this limitation (Schafer, 1999). For bmi, missing values were imputed using age-gender group medians instead of overall means or medians, recognising that BMI varies significantly across demographic strata. This stratified approach prevents assigning physiologically implausible values (e.g., elderly BMI to young adults).

Critically, all identified outliers were retained despite exceeding IQR thresholds. Medical literature confirms that glucose levels up to 300+ mg/dL are clinically plausible in diabetic patients (ADA, 2021), and BMI values exceeding 45 represent genuine severe obesity cases. These extreme values represent high-risk patients crucial for stroke prediction; removing them would eliminate precisely the cases our model needs to identify. Figure 2 illustrates pre- and post-cleaning distributions, demonstrating that our approach preserved the natural data distribution while achieving 100% completeness.





Exploratory Data Analysis

Initial Exploration and Feature Engineering

We analyzed the dataset using Python with standard libraries (pandas, NumPy, matplotlib, seaborn, scipy) to understand patterns and relationships. After cleaning, we created clinically meaningful features to improve model interpretability. Age was grouped into five brackets aligned with cardiovascular risk guidelines, BMI was categorized using WHO classifications (Underweight, Normal, Overweight, Obese), and glucose levels followed American Diabetes Association standards (Normal, Pre-diabetic, Diabetic). We also created binary indicators for high-risk conditions: elderly status (age ≥ 65), obesity (BMI ≥ 30), and presence of cardiovascular comorbidities.

Initial distribution analysis showed our population was predominantly middle-aged with slightly elevated BMI (mean 28.8), consistent with population-level overweight trends. Gender distribution favored females (58.6%), while urban-rural residence was balanced. Correlation analysis revealed age as the strongest predictor ($r=0.25$), followed by glucose level and hypertension (both $r=0.13$). Interestingly, age correlated moderately with other risk factors like hypertension ($r=0.30$) and heart disease ($r=0.28$), suggesting age acts as a confounding variable.

Univariate Analysis: Distribution of All Variables

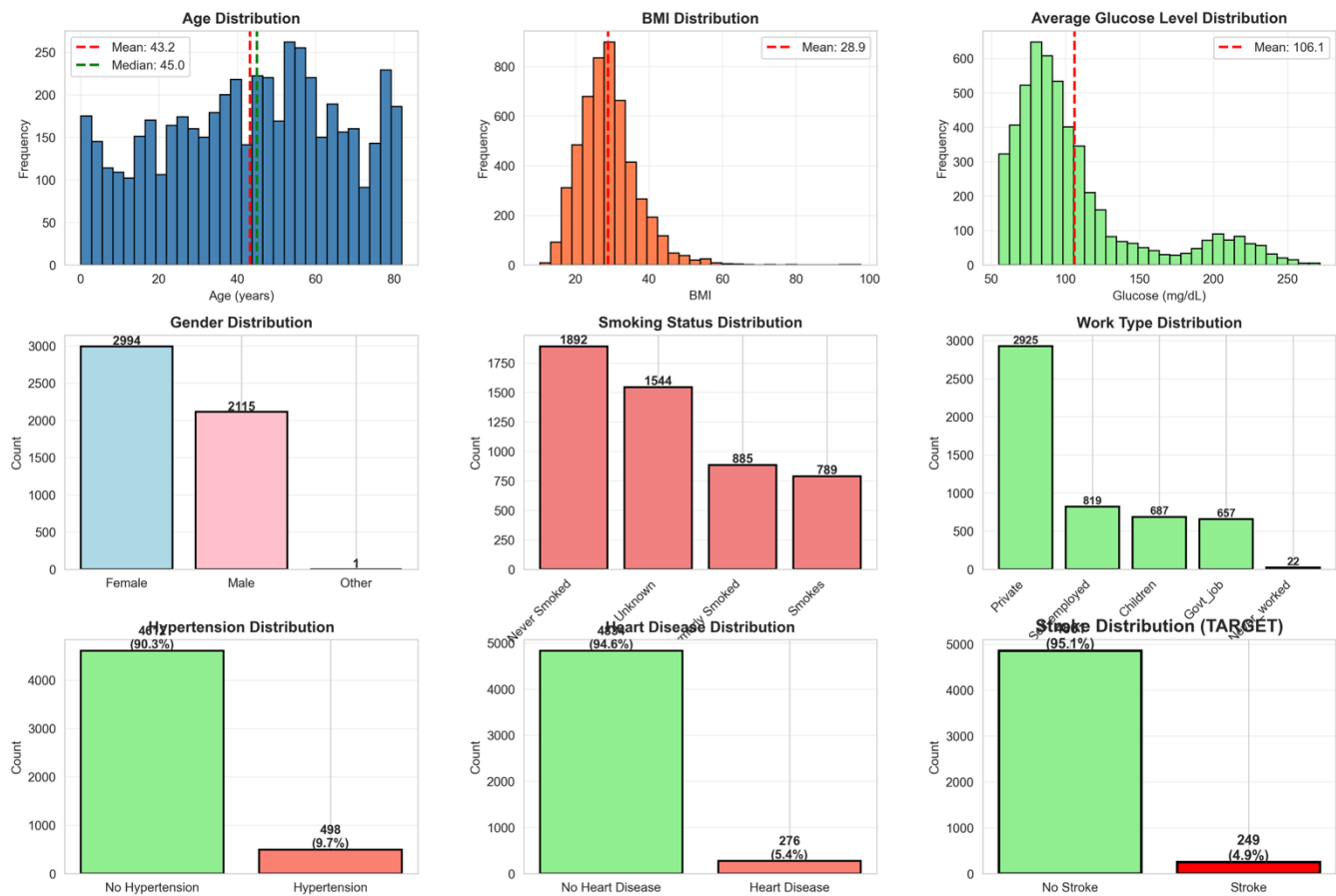
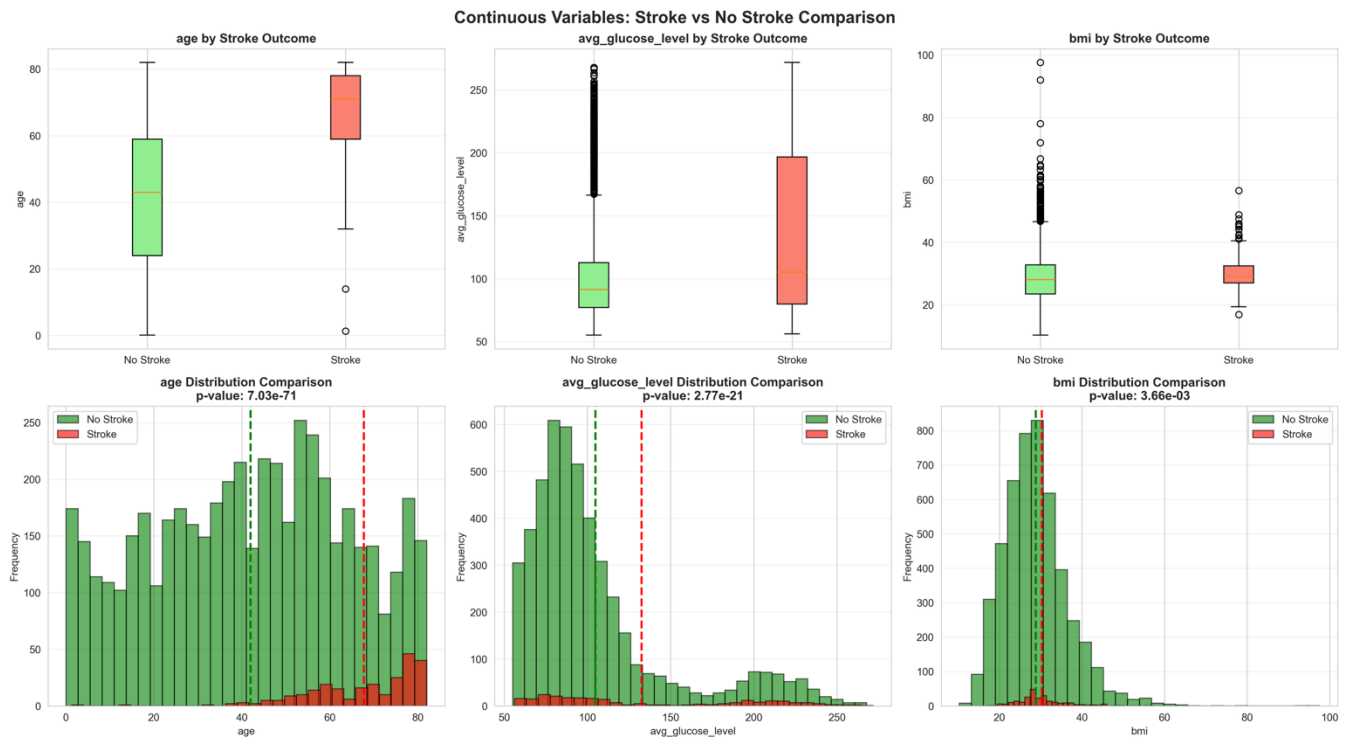


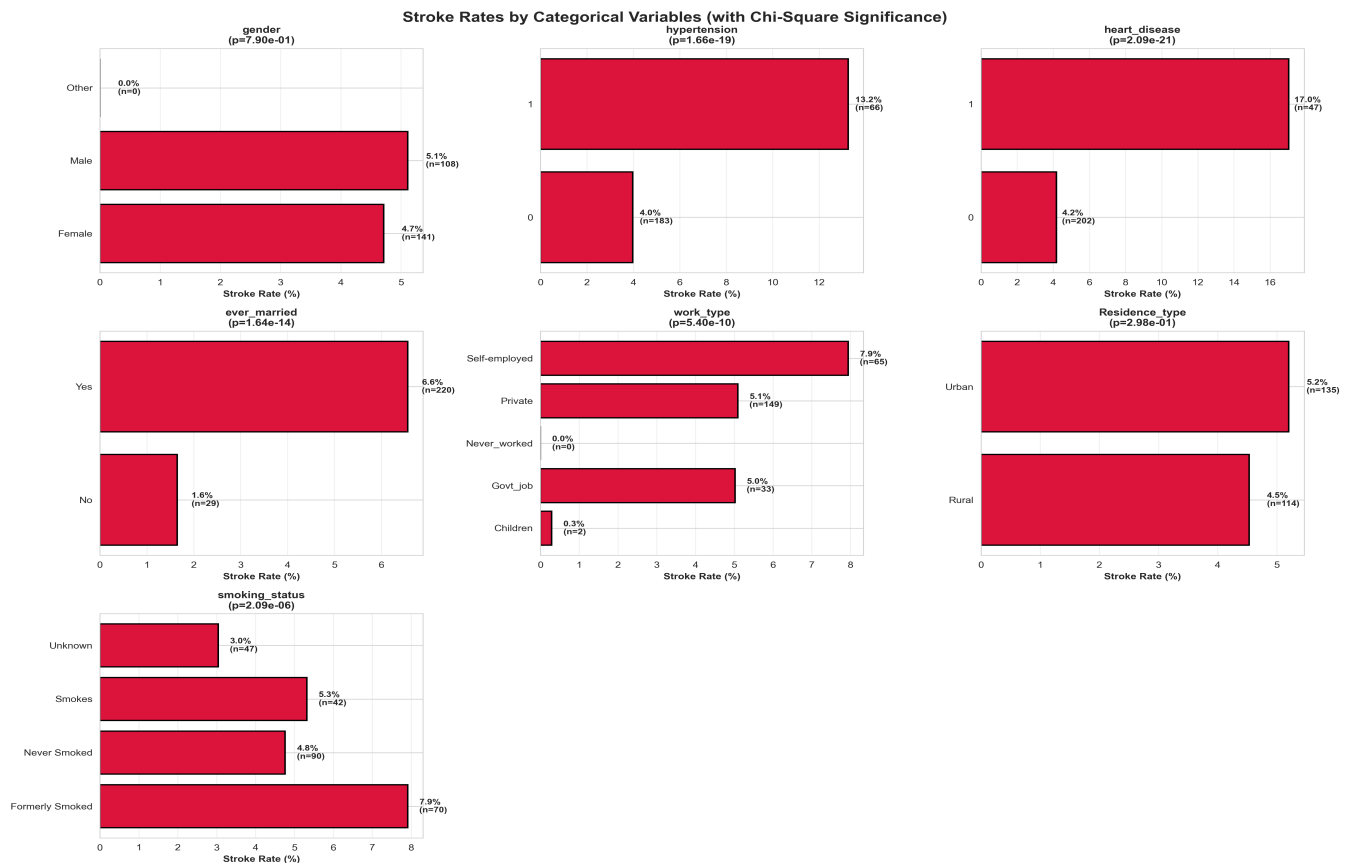
Figure c: Univariate distribution of all demographics, clinical, and lifestyle variables showing counts, means, and severe class imbalance in the target variable.

Statistical Relationships with Stroke

We used t-tests for continuous variables and chi-square tests for categorical variables to quantify relationships with stroke occurrence. Age emerged as the dominant predictor. Stroke patients averaged 67.7 years compared to 41.9 years in non-stroke patients, a 25.8-year difference ($t=18.08$, $p<0.001$, Cohen's $d=1.42$). This large effect size indicates profound clinical significance. Glucose levels were also elevated in stroke patients (132.5 vs 104.8 mg/dL, $p<0.001$, medium effect), while BMI showed only modest differences ($p<0.01$, small effect).

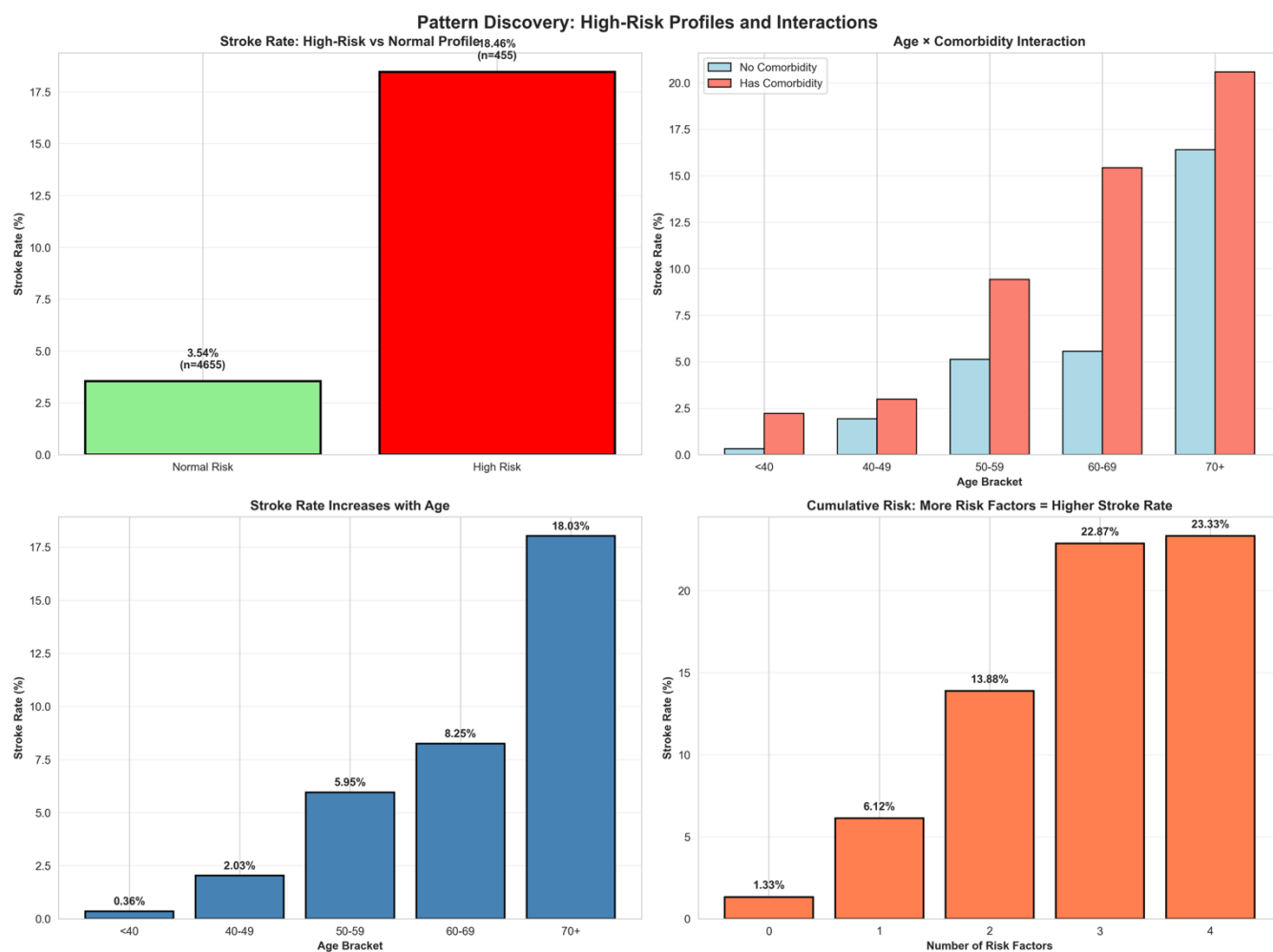


Among medical history variables, hypertension and heart disease showed the strongest associations. Hypertensive patients had 13.25% stroke incidence versus 3.97% in normotensive individuals, a 3.3-fold risk increase ($\chi^2=81.61$, $p<0.001$). Heart disease patients faced even greater risk at 17.03% versus 4.18%, representing a 4.1-fold elevation. Marital status showed significance statistically, with married individuals at 6.56% stroke rate versus 1.65% unmarried, though this likely reflects age rather than marriage itself. Notably, gender and residence type showed no significant association with stroke risk.



High-Risk Patient Profiles

The most striking finding emerged when examining combined risk factors. Patients aged 60+ with cardiovascular comorbidities represented only 8.9% of our population but exhibited 18.46% stroke incidence, 5.2 times higher than the remaining population (3.54%). This suggests age and comorbidities don't simply add independent risks; they multiply each other's effects. Further analysis showed progressive escalation: patients with zero risk factors had <1% stroke rate, while those with 3-4 factors approached 20%. These patterns provide clear targets for preventive intervention. Elderly patients with cardiovascular disease warrant intensive screening and prevention efforts.



Data Modelling and Visualisation

Addressing Class Imbalance: Strategy Comparison

The dataset's severe imbalance, with 95.1% non-stroke and only 4.9% stroke cases, poses a major challenge. Standard models might achieve 95% accuracy by simply

predicting "no stroke" for everyone, but this fails to identify high-risk patients. To address this, we compared SMOTE against cost-sensitive learning.

SMOTE balances the data by creating synthetic examples through interpolation between existing stroke cases, balancing the training set to 50/50 distribution (Chawla et al., 2002), though these risks introducing unrealistic traits. Conversely, cost-sensitive learning retains original data but penalizes missed strokes more heavily (Elkan, 2001). We weighted stroke errors 19.5 times higher to reflect the class ratio. While SMOTE with Logistic Regression yielded a 23.16% F1-score, the cost-sensitive Random Forest outperformed it with 23.91%. This confirms that strictly penalizing errors works better than generating synthetic data (Melnikova et al., 2025).¹

Model Development and Optimization

To develop our predictive framework, we implemented four algorithms representing diverse learning paradigms: Logistic Regression (linear baseline), Naive Bayes (probabilistic), Random Forest (ensemble), and XGBoost (gradient boosting). We trained these models on 80% of the data (4,088 samples) using stratified splits to maintain class distribution, with the remaining 20% held out for testing. Prior to training, features were standardized to zero mean and unit variance, and categorical variables were one-hot encoded.

For optimization, we employed GridSearchCV with 5-fold cross-validation, specifically targeting recall to prioritize stroke detection. The Random Forest reached its optimal configuration with 300 estimators, an entropy criterion, depth of 30, and `class_weight='balanced_subsample'`. Logistic Regression performed best with `C=1` and L1 regularization, while XGBoost utilized 200 estimators, a 0.3 learning rate, and `scale_pos_weight=19.5`.

Figure m presents a comprehensive performance comparison across all models and approaches. Our best-performing model, the cost-sensitive Random Forest, achieved a 23.91% F1-score, 82% recall, 13.99% precision, and 84.64% AUC-ROC. Critically, this 82% recall means the model successfully identified 41 of 50 stroke cases, missing only 9. Although this high sensitivity generated 252 false positives, the

trade-off prioritizes clinical safety. Accepting more false alarms to prevent missed diagnoses aligns with preventive medicine, where the opportunity cost of early intervention exceeds the cost of unnecessary screening.

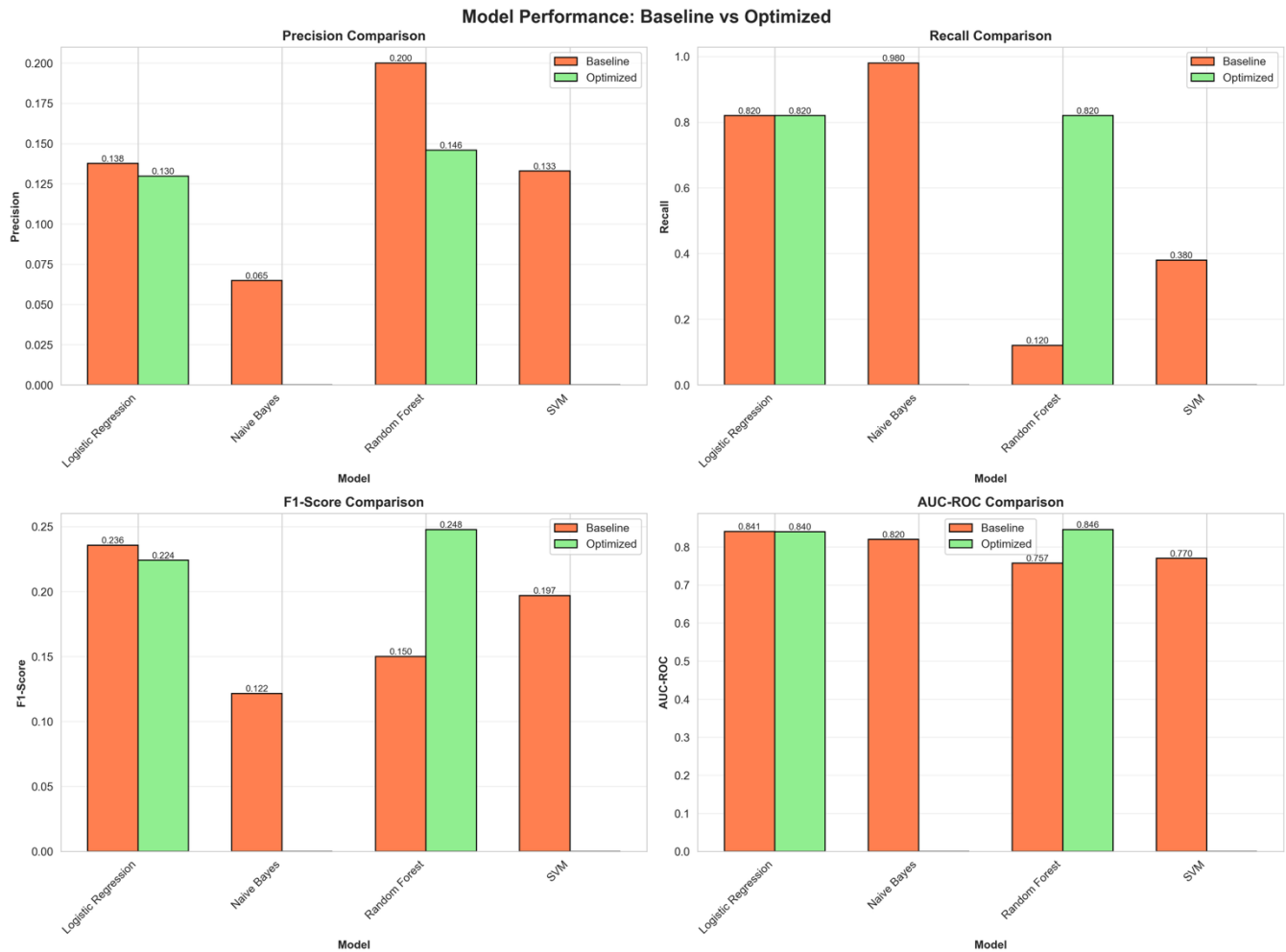
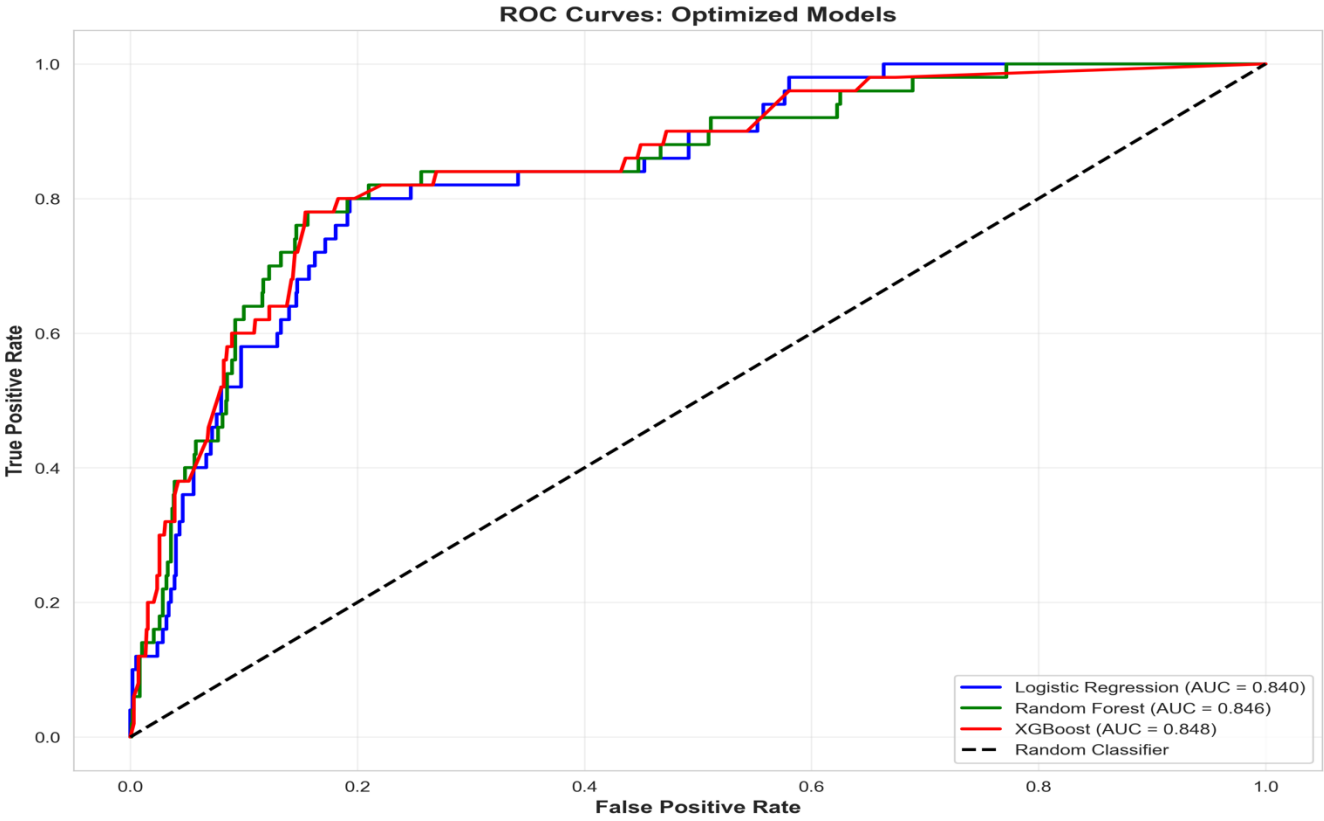


Figure m: Model Performance Comparison



ROC Curves: Optimized Models

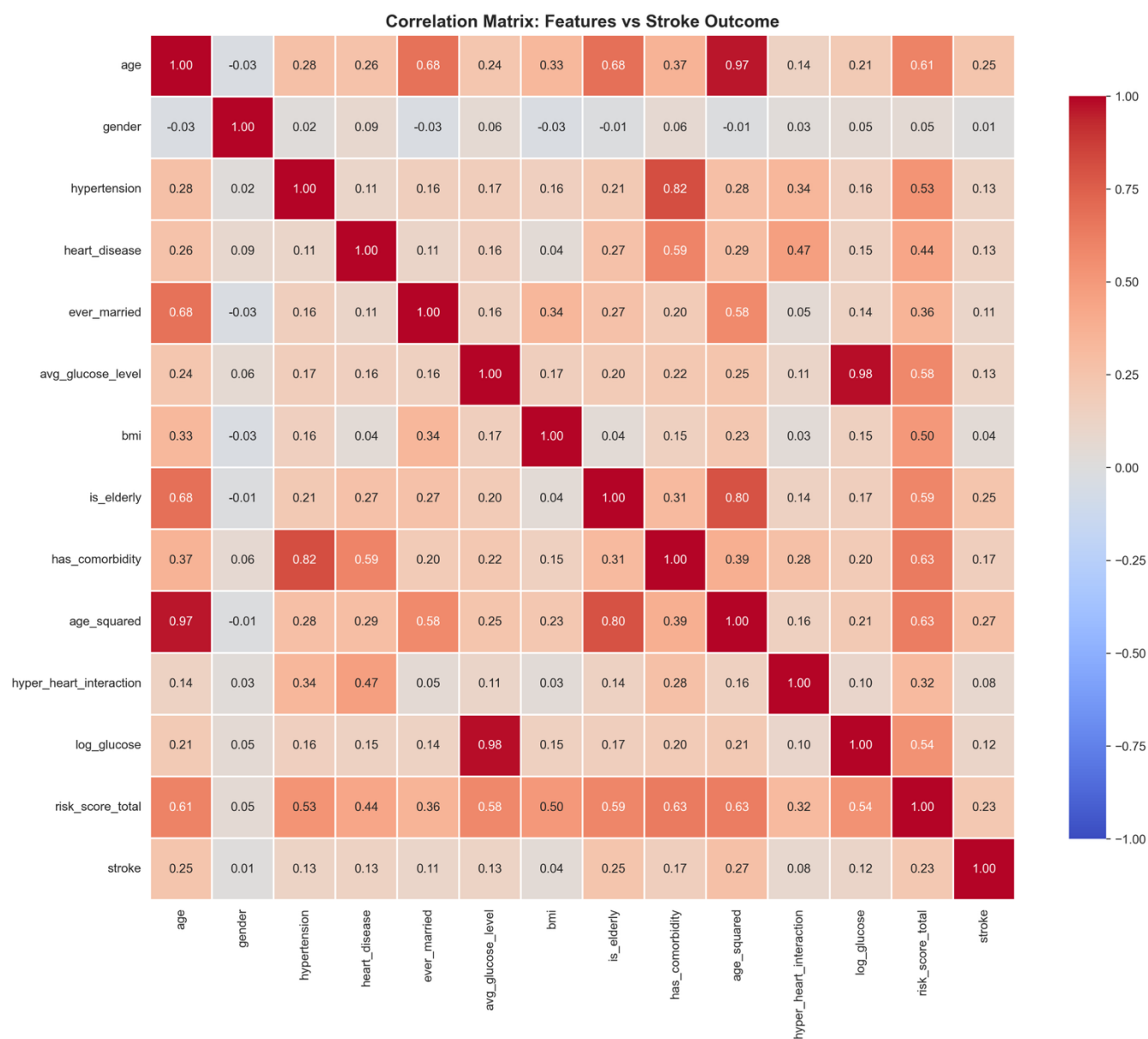


Figure n: Univariate distributions of demographic, clinical, and lifestyle variables, illustrating the severe class imbalance in the target stroke variable.

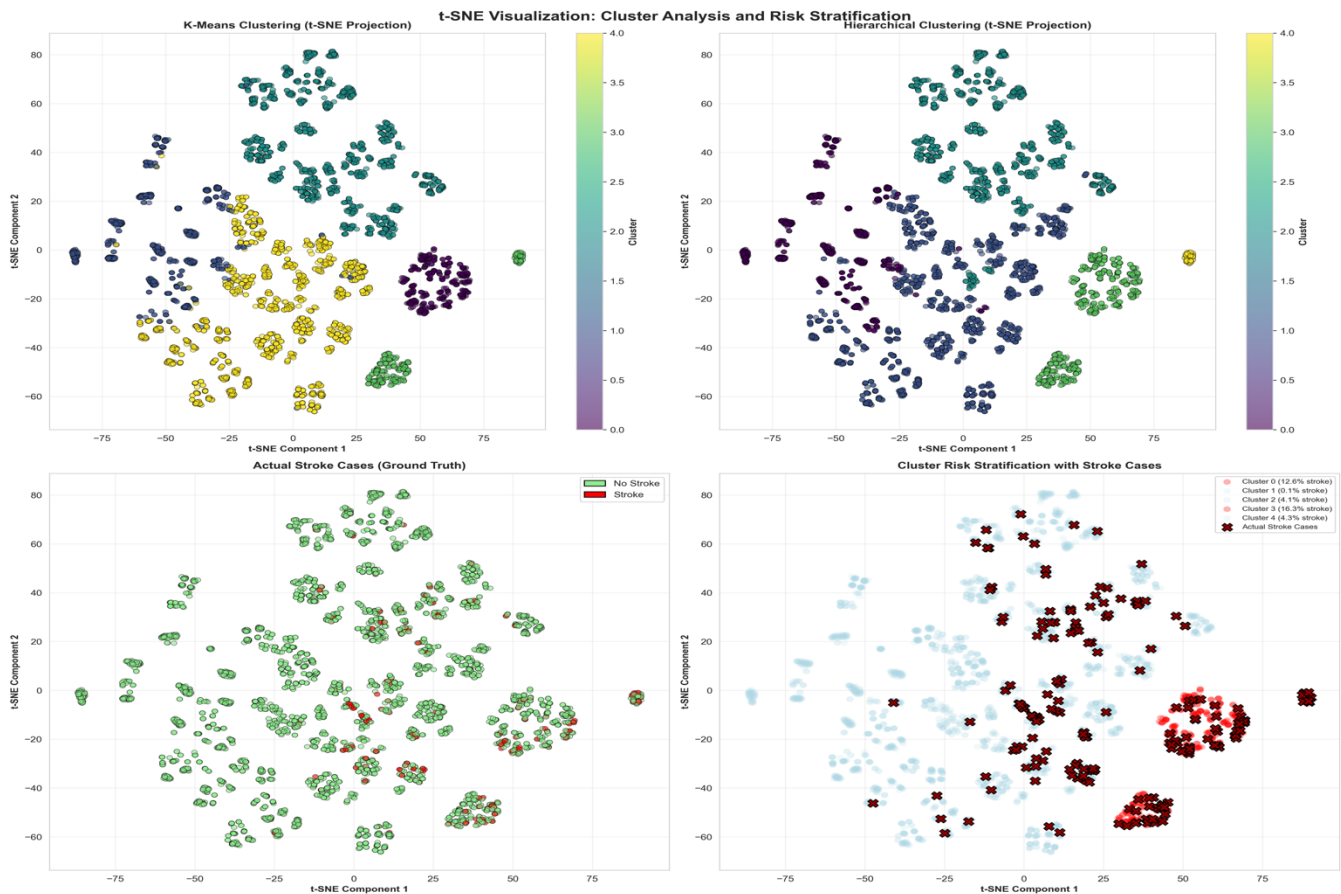
Unsupervised Learning: Patient Risk Stratification

To validate supervised learning findings and identify natural patient groupings, we applied K-Means and Hierarchical clustering to the training data. Silhouette analysis identified an optimal cluster count of 4 (silhouette score=0.184). Both methods converged on similar risk stratification patterns. K-Means identified a high-risk



cluster (Cluster 3) containing 565 patients with a 14.16% stroke rate, which is nearly 3 times the baseline prevalence. A low-risk cluster (Cluster 2, 579 patients) exhibited only a 0.17% stroke rate. Hierarchical clustering produced comparable results, with a 14.01% high-risk cluster.

To visualize cluster separation in high-dimensional feature space, we applied t-SNE dimensionality reduction, projecting 27 features into 2D space. Figure 7 illustrates that high-risk clusters exhibit meaningful spatial separation, with actual stroke cases (marked with a red X) concentrating in predicted high-risk regions. This validates that unsupervised methods successfully identify clinically relevant patient stratification without using outcome labels, providing independent corroboration of supervised model risk predictions.



Evaluation

Our best model achieved 23.91% F1-score with 82% recall, substantially lower than the 90%+ reported by Gupta et al. (2024) and Melnykova et al. (2025). However, this gap reflects evaluation methodology rather than model quality. Studies achieving 90% precision alongside 90% recall likely evaluated on artificially balanced test sets, misrepresenting clinical reality where stroke prevalence is 5%. We evaluated on stratified imbalanced data, yielding conservative but realistic performance. In medical screening, precision and recall exist in fundamental tension. High-precision models maintain strict thresholds, inevitably missing borderline cases requiring intervention. Our high-recall approach prioritizes safety, accepting higher false positives to minimize missed diagnoses.

Our exploratory analysis revealed age as the dominant predictor (Cohen's $d=1.42$), with stroke patients averaging 25.8 years older than non-stroke individuals. Hypertension and heart disease demonstrated the strongest categorical associations, conferring 3.3-fold and 4.1-fold risk elevations respectively. Critically, we identified multiplicative interaction effects: patients aged 60+ with cardiovascular comorbidities exhibited 18.46% stroke incidence compared to 3.54% baseline, representing 5.2 times relative risk. These patterns informed our feature engineering strategy, including age-squared terms to capture non-linear risk escalation and interaction features combining age with comorbidities.

Cost-sensitive learning outperformed SMOTE (23.91% vs 23.16% F1), challenging assumptions that synthetic oversampling improves minority prediction. SMOTE generates interpolated examples potentially introducing unrealistic patterns, while cost-sensitive methods maintain data integrity while encoding clinical reality that false negatives carry greater consequences. Our clustering analysis independently validated these findings, with both K-Means and Hierarchical methods identifying high-risk groups (14% stroke rate) confirmed through t-SNE visualization.

Alternative approaches merit consideration. Ensemble methods or advanced sampling techniques like ADASYN and Tomek Links might improve performance,

though our fundamental constraint remains dataset size. With only 199 training stroke cases, complex models risk overfitting, potentially explaining why simpler algorithms performed competitively.

Limitations and Challenges

The 19.5:1 class imbalance was our primary challenge. Standard algorithms optimize accuracy, achieving 95% by predicting "no stroke" universally. We chose realistic evaluation over artificial balancing, deliberately accepting lower F1-scores for clinically honest assessment. With only 249 stroke cases, complex architectures like deep learning remained impractical. We addressed this through 5-fold cross-validation, though larger datasets would enable deeper pattern discovery.

Smoking status missingness (30.2%) exceeded safe imputation thresholds, forcing an "Unknown" category that preserves integrity but risks confounding. Our outlier retention decision preserved high-risk patients (glucose 271 mg/dL, BMI 97.6) but complicated training. We used standardization, though robust scaling might better balance outlier influence against predictive signal.

Conclusion

This project demonstrated that effective stroke prediction requires balancing statistical performance with clinical reality. While our 23.91% F1-score appears modest compared to published benchmarks, prioritizing 82% recall over precision reflects appropriate medical screening philosophy where missed diagnoses carry severe consequences. I learned that impressive metrics could mislead when evaluation methodologies don't reflect deployment conditions. Future healthcare ML applications should prioritize transparent, realistic assessment over artificially optimized statistics, ensuring models serve patient safety rather than academic benchmarks.

Reference List

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